

THE LEFSCHETZ DICHOTOMY: WHY THE HARD-LEFSCHETZ ROUTE TO THE RIEMANN HYPOTHESIS IS OBSTRUCTED BY LINEAR INDEPENDENCE

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ABSTRACT. Weil’s proof of the Riemann Hypothesis for curves over finite fields is an intersection form plus a Hodge index: positivity on the primitive cohomology, delivered by an ample class (a hard-Lefschetz $\mathfrak{sl}_2$ -action). Transcribing the construction to $\mathrm{Spec} \mathbb{Z}$ produces, unconditionally, every ingredient except that ample class. We make the missing object precise and prove that it cannot be supplied along this route. First we record an *independence filter*: any raising operator built as a function of the Frobenius flow $\mathcal{T}$ (whose spectrum is the zeros) only restates the Weil positivity. Then we construct the candidate Kähler data from the archimedean place: the density $\Psi(r) = \mathrm{Re} \psi(\frac{1}{4} + \frac{ir}{2}) - \log \pi$ is the smooth zero-counting density; it is strictly increasing on $r > 0$ with a unique sign change $r_0 = 6.2898 \dots$ lying below the first zero, hence positive on the spectrum but indefinite on the low band. We show the resulting form is indefinite on the primitive part, exhibit an explicit hard-Lefschetz $\mathfrak{sl}_2$ on the archimedean continuum whose raising operator is provably outside the von Neumann algebra of $\mathcal{T}$ — the first such operator this line of work produces — and verify that the transport to the discrete zeros is exactly the explicit formula, with the Weil positivity emerging only as the residual of an almost total archimedean–prime cancellation. Finally we prove the *Lefschetz dichotomy*: on the zeros the linear independence of the ordinates forbids any Tate-twist raising operator, while on the continuum the only available $\mathfrak{sl}_2$ is principal series and carries no Hodge–Riemann form. Thus an arithmetic spectrum and a Hodge–Riemann-carrying grading are linearly-independence-incompatible. This is a theorem about why ζ resists the Lefschetz route, sharpening the finitization-under-LI obstruction of the series. All claims are accompanied by reproducible computations; none asserts a proof of RH.

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1. INTRODUCTION

Weil's proof of RH for a curve $C/\mathbb{F}_q$ has two parts: an intersection pairing on the surface $C \times C$, and the Hodge index theorem, which makes the pairing negative definite on the primitive part. The second part is supplied by an *ample class* on the surface — equivalently a hard-Lefschetz $\mathfrak{sl}_2$ -action (L, Λ, H) whose positivity is the Hodge–Riemann bilinear relation. The Frobenius acts on H^1 , whose eigenvalues are the zeros, and the ample class is independent geometric input.

The program of which this paper is a part has built, unconditionally and with numerical verification, the analogue of the *first* part for $\text{Spec } \mathbb{Z}$: a Kreĭn space $\mathcal{H}_W$ with an intersection pairing whose negative index counts the off-line zeros, a Lefschetz trace identity (the explicit formula), a Rankin–Selberg effectivity, and a positive ample cone [1, 4, 5]. The Riemann Hypothesis is then equivalent to the *second* part: positivity of the pairing on the primitive subspace — the arithmetic Hodge index. Eight successive sharpenings reduced the gap to a single missing object: an independent, positive, hard-Lefschetz $\mathfrak{sl}_2$ -action (a Kähler class ω) on the primitive cohomology, commuting appropriately with the Frobenius flow $\mathcal{T}$ whose spectrum is the set of ordinates $\{\gamma_\rho\}$.

This paper does two things. It records why the obvious construction — building ω from the spectral data $(\mathcal{T}, J, Q)$ — can only restate RH (§2). And it carries out the one construction that escapes this circularity, the archimedean one, to its conclusion (§§3–6), proving along the way the first hard-Lefschetz raising operator independent of $\mathcal{T}$ (§5), and ending at a clean structural obstruction: the *Lefschetz dichotomy* (§7). The dichotomy explains, in representation-theoretic terms, why the linear independence of the zeros (the hypothesis LI, that the positive ordinates are $\mathbb{Q}$ -linearly independent) blocks the Lefschetz route. Nothing here proves or assumes RH; the result is negative and structural, and every quantitative claim is reproducible from the code in Appendix A.

2. THE INDEPENDENCE FILTER

We work on the primitive part $\mathcal{H}_W^{\text{prim}}$, an infinite-dimensional real Kreĭn space carrying: the indefinite Weil pairing Q (negative index = number of off-line zeros), a self-adjoint Frobenius operator $\mathcal{T}$ with $\text{spec}(\mathcal{T}) = \{\gamma_\rho\}$, and a complex structure J from the spectral splitting [1]. Let $W^*(\mathcal{T})$ denote the von Neumann algebra generated by $\mathcal{T}$.

Proposition 2.1 (Determinacy / the filter). *The only $(1, 1)$ -form compatible with (J, Q) in the standard way is $\omega(x, y) = Q(Jx, y)$, and its positivity $\omega(x, Jx) = Q(Jx, Jx) > 0$ is exactly RH. Consequently, if a raising operator L lies in $W^*(\mathcal{T})$ — i.e. $L = f(\mathcal{T})$ for some Borel f — then the associated form $Q(Jf(\mathcal{T})\cdot, \cdot)$ is a function of the spectral data and asserting its positivity restates RH. An ω that forces positivity must use a raising operator $L \notin W^*(\mathcal{T})$.*

Proof. Compatibility of a $(1, 1)$ -form with (J, Q) means $\omega(x, y) = Q(Jx, y)$; this fixes ω . Its positivity is the second Riemann bilinear relation $Q(Jx, Jx) > 0$, equivalent by the Weil criterion to the vanishing of the negative index, i.e. RH. If $L = f(\mathcal{T})$ then L commutes with $\mathcal{T}$ and is diagonal in the spectral resolution of $\mathcal{T}$; every quantity built from L, J, Q is a function of that resolution, so its sign is the sign of Q on the spectral subspaces — the statement RH itself. $\square$

The filter is the lesson of the eight attempts in compact form: a candidate Kähler class is genuine input only if its raising operator is *not* a function of the Frobenius. We now produce one.

3. THE ARCHIMEDEAN DENSITY

The archimedean place contributes to the Weil quadratic form, for a test function f with Fourier transform $\hat{f}$, the term

$$(1) \quad A_\infty(f, f) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(r)|^2 \Psi(r) dr, \quad \Psi(r) = \text{Re } \psi\left(\frac{1}{4} + \frac{ir}{2}\right) - \log \pi,$$

where $\psi = \Gamma'/\Gamma$ and Ψ is the smooth density of the archimedean factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ in the explicit formula. The following structure is elementary but decisive.

Proposition 3.1 (Structure of Ψ). *Ψ is even and real-analytic, and:*

- (i) $\Psi'(r) = \sum_{n \geq 0} \frac{(\frac{1}{4} + n)r}{((\frac{1}{4} + n)^2 + (r/2)^2)^2} > 0$ for $r > 0$ (strictly increasing);
- (ii) Ψ has a unique positive zero $r_0 = 6.2898359888\dots$, global minimum $\Psi(0) = \psi(\frac{1}{4}) - \log \pi = -\gamma - 3 \log 2 - \frac{\pi}{2} - \log \pi = -5.37218\dots$;
- (iii) $\Psi(r) = \log \frac{|r|}{2\pi} + o(1)$, and $\frac{1}{2\pi}\Psi(r)$ equals the Riemann–von Mangoldt zero density $N'(r)$ to leading order.

In particular $\Psi(r) > 0 \iff |r| > r_0$; the only negativity is the bounded band $|r| < r_0$, which lies strictly inside the zero-free gap, $r_0 < \gamma_1 = 14.1347\dots$, so $\Psi > 0$ on the entire support of the zeros.

Proof. Evenness and analyticity are clear. For (i), differentiate (1): $\Psi'(r) = \operatorname{Re}[\frac{i}{2}\psi'(\frac{1}{4} + \frac{ir}{2})] = -\frac{1}{2}\Im\psi'(\frac{1}{4} + \frac{ir}{2})$, and $\psi'(z) = \sum_{n \geq 0} (z + n)^{-2}$ gives $\Im\psi'(\frac{1}{4} + \frac{ir}{2}) = -\sum_n \frac{2(\frac{1}{4} + n)(r/2)}{((\frac{1}{4} + n)^2 + (r/2)^2)^2} < 0$ for $r > 0$; the stated series follows. Hence Ψ increases from $\Psi(0) < 0$ to $+\infty$, crossing zero once at r_0 ; (ii)–(iii) are the standard values and the digamma asymptotics $\operatorname{Re}\psi(\frac{1}{4} + \frac{ir}{2}) = \log \frac{|r|}{2} + o(1)$. The numerical constants and $r_0 < \gamma_1$ are verified to 40 digits in Appendix A. $\square$

So the archimedean form is a strictly positive metric on the spectral bulk ($|r| \geq \gamma_1$, where $\mathcal{T}$ has spectrum) and indefinite only on a finite band in the spectral gap.

4. INDEFINITENESS ON THE PRIMITIVE PART

A natural hope is that the band $|r| < r_0$ is absorbed by the ample/pole part of the construction (the de Vallée Poussin edge positivity of [4]). It is not.

Proposition 4.1 (A_{∞} is indefinite on primitives). *The vectors $\hat{f}_{\sigma}(r) = (r^2 + \frac{1}{4})e^{-r^2/2\sigma^2}$ are primitive ($\hat{f}_{\sigma}(\pm \frac{i}{2}) = 0$, so the pole/ample term vanishes on them), and $A_{\infty}(f_{\sigma}, f_{\sigma}) < 0$ for small σ : numerically $A_{\infty}/\|f_{\sigma}\|^2 = -1.82$ at $\sigma = 1$, crossing to positive near $\sigma \approx 5$. Hence A_{∞} is indefinite on $\mathcal{H}_W^{\text{prim}}$ and the band is not absorbed by the pole part.*

Proof. $(i/2)^2 + \frac{1}{4} = 0$ gives $\hat{f}_{\sigma}(\pm i/2) = 0$, so f_{σ} is primitive and the pole term $2|\hat{f}_{\sigma}(i/2)|^2$ vanishes identically. For small σ the mass of $|\hat{f}_{\sigma}|^2$ concentrates in $|r| < r_0$ where $\Psi < 0$, giving $A_{\infty} < 0$; the values are computed in Appendix A. $\square$

The signed form factors through a positive metric and a grading sign: $A_{\infty} = g_{\infty}(S \cdot, \cdot)$ with $g_{\infty} = |\Psi|(\mathcal{T}_{\text{cont}}) \succ 0$ and $S = \operatorname{sgn}(\mathcal{T}_{\text{cont}}^2 - r_0^2) \in \{\pm 1\}$ — the shape of a Hodge–Riemann form, a positive metric times a degree sign. The indefiniteness is a candidate grading, not a defect; whether it organizes into a hard-Lefschetz $\mathfrak{sl}_2$ is the next question.

5. THE CONTINUUM $\mathfrak{sl}_2$ AND THE DISCRETE NO-GO

The Frobenius $\mathcal{T}$ acts on $\mathcal{H}_W$ with simple spectrum, so its commutant is a maximal abelian subalgebra (a MASA): any L commuting with $\mathcal{T}$ is a function of $\mathcal{T}$ and, by Proposition 2.1, useless. The correct hard-Lefschetz condition is therefore not commutation but the *Tate twist* $[\mathcal{T}, L] = cL$ — the analogue of $\operatorname{Frob} \cdot L = q L \cdot \operatorname{Frob}$ on $C \times C$, where L raises the weight and scales the Frobenius eigenvalue.

Proposition 5.1 (No Tate-twist Lefschetz on the zeros). *If $[\mathcal{T}, L] = cL$ with $c \neq 0$ and $L \neq 0$, then L maps the γ -eigenspace of $\mathcal{T}$ into the $(\gamma + c)$ -eigenspace, so the zero set must satisfy $\{\gamma_\rho\} + c \subseteq \{\gamma_\rho\}$, i.e. be invariant under translation by c . No $c \neq 0$ has this property: a single inclusion $\gamma, \gamma + c, \gamma + 2c \in \{\gamma_\rho\}$ already is a nontrivial $\mathbb{Q}$ -linear relation among ordinates, excluded by LI [3]; unconditionally, the variable spacing of the zeros rules out exact translation invariance. Hence no hard-Lefschetz raising operator exists on $\mathcal{H}_W$ itself.*

The remedy is to pass to the archimedean continuum $L^2(\mathbb{R}, dr)$, where the multiplication operator $\mathcal{T}_{\text{cont}}$ has spectrum all of $\mathbb{R}$ — which is translation invariant.

Proposition 5.2 (Explicit continuum $\mathfrak{sl}_2$). *On $L^2(\mathbb{R}, dr)$ put $L = T_c$ (translation by c), $H = \frac{2}{c} \mathcal{T}_{\text{cont}}$, and $\Lambda = T_{-c} m(\mathcal{T}_{\text{cont}})$ with $m(r) = -\frac{r^2}{c^2} + \frac{r}{c}$. Then*

$$[\mathcal{T}_{\text{cont}}, L] = cL, \quad [H, L] = 2L, \quad [H, \Lambda] = -2\Lambda, \quad [L, \Lambda] = H,$$

so (L, Λ, H) is an $\mathfrak{sl}_2$ -triple, Tate-twist compatible with the Frobenius flow. The raising operator $L = T_c$ is not a multiplication operator, hence $L \notin W^*(\mathcal{T}_{\text{cont}})$: the independence filter of Proposition 2.1 is cleared by an explicit operator. The grading $H = \frac{2}{c} \mathcal{T}_{\text{cont}}$ has continuous spectrum — a “Frobenius-as-a-flow” Lefschetz, not an integer grading.

Proof. $[\mathcal{T}_{\text{cont}}, T_c] = cT_c$ is the canonical commutation of multiplication and translation. Then $[H, L] = \frac{2}{c} [\mathcal{T}_{\text{cont}}, T_c] = 2T_c = 2L$. With $(T_{-c} m T_c f)(r) = m(r + c)f(r)$ one computes $[L, \Lambda] = [T_c, T_{-c} m] = m(r) - m(r + c) = \frac{2}{c} r = H$ (using $m(r) - m(r + c) = -\frac{1}{c^2}(r^2 - (r + c)^2) + \frac{1}{c}(r - (r + c)) = \frac{2}{c} r$), and $[H, \Lambda] = \frac{2}{c} [\mathcal{T}_{\text{cont}}, T_{-c} m] = -2\Lambda$. The relations are verified numerically to 10^{-12} in Appendix A. A translation is not diagonal in the position representation, so $L \notin W^*(\mathcal{T}_{\text{cont}})$. $\square$

Propositions 5.1–5.2 are the two horns to be joined in §7. First we check what the construction buys downstairs.

6. THE TRANSPORT IS THE EXPLICIT FORMULA

The continuum structure connects to the arithmetic zeros only through sampling.

Proposition 6.1 (Transport identity). *For primitive f the explicit formula reads*

$$(2) \quad Q(f, f) = \sum_{\rho} |\widehat{f}(\gamma_\rho)|^2 = A_\infty(f, f) - P(f, f), \quad P(f, f) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n),$$

where g is the Fourier transform of $|\widehat{f}|^2$ (the pole term drops since $\widehat{f}(\pm i/2) = 0$). On the test family of Proposition 4.1 the identity holds to 10^{-12} – 10^{-20} , and A_∞ and P are large, negative, and nearly equal, so Q is the small non-negative residual of an almost total archimedean–prime cancellation.

Table 1 records the computation (first 300 zeros, von Mangoldt to $2 \cdot 10^4$). The point is structural: the discrete positivity $Q \geq 0$ is *not* the transported image of the upstairs metric — on these primitives $A_\infty < 0$ — but the residue $A_\infty - P$, the classical wrong-sign obstruction [2].

σ	$Q = \sum_{\rho} \widehat{f}(\gamma_\rho) ^2$	A_∞	P	$Q - (A_\infty - P)$
2	1.6×10^{-17}	-6.2225	-6.2225	-1.4×10^{-20}
3	1.83×10^{-5}	-21.677	-21.677	-2×10^{-12}
4	0.3022	-24.239	-24.542	-3×10^{-4}

TABLE 1. The transport identity (2) and the near-cancellation.

7. THE LEFSCHETZ DICHOTOMY

The continuum $\mathfrak{sl}_2$ of Proposition 5.2 clears the independence filter, but it does not carry the positivity that hard Lefschetz needs.

Proposition 7.1 (No Hodge–Riemann form on the continuum $\mathfrak{sl}_2$). *The representation of Proposition 5.2 is principal series: $\text{spec}(H) = \frac{2}{c}\mathbb{R}$ is unbounded below, and the lowest-weight equation $\Lambda v = 0$ forces $m(r)v(r) = 0$, i.e. v supported on the measure-zero set $\{0, c\}$, so there is no normalizable lowest-weight vector and no primitive decomposition $V = \bigoplus_k L^k(\ker \Lambda)$. Hodge–Riemann bilinear relations are defined only on lowest-weight (equivalently finite-dimensional or discrete-series, integer-graded) $\mathfrak{sl}_2$ -representations; the continuum $\mathfrak{sl}_2$ carries none. Concretely, the only manifestly positive form the data yield is $g_\infty(f, f) = \frac{1}{2\pi} \int |\hat{f}|^2 |\Psi|$, which differs from the Weil form Q by factors 10^2 – 10^{18} on the test family (Appendix A); the degree grading produces g_∞ , not Q .*

Theorem 7.2 (The Lefschetz dichotomy). *Under LI there is no hard-Lefschetz $\mathfrak{sl}_2$ -action carrying a Hodge–Riemann form that acts on the arithmetic zero-spectrum:*

discrete horn: on $\{\gamma_\rho\}$ no Tate-twist raising operator exists (Prop. 5.1),
continuum horn: on $L^2(\mathbb{R}, dr)$ the raising operator exists but the $\mathfrak{sl}_2$
 is principal series, with no Hodge–Riemann form (Prop. 7.1).

Equivalently: a Hodge–Riemann-carrying grading requires lowest weight (integer spacing), which forces the Frobenius eigenvalues into a single weight; LI is exactly the statement that the zeros admit no such integer-weight structure. An arithmetic spectrum and a Hodge–Riemann grading are LI-incompatible.

Proof. The two horns are Propositions 5.1 and 7.1. For the equivalence: in the function-field case H^1 is pure of weight 1, the Frobenius eigenvalues are Weil numbers of that single weight, and the Lefschetz $\mathfrak{sl}_2$ on $H^0 \oplus H^1 \oplus H^2$ is lowest weight, so Hodge–Riemann applies and yields RH. A lowest-weight (integer-graded) $\mathfrak{sl}_2$ commuting-up-to-Tate-twist with $\mathcal{T}$ would, by the argument of Proposition 5.1, place the eigenvalues of $\mathcal{T}$ in an arithmetic progression of weights — a $\mathbb{Q}$ -linear structure on $\{\gamma_\rho\}$ excluded by LI. Hence no such grading exists on the zeros; buying one on the continuum costs the lowest-weight property (Proposition 7.1). $\square$

Remark 7.3. Theorem 7.2 refines the finitization-under-LI obstruction of [3] from “no finite model” to a representation-theoretic statement about gradings, and it is reached constructively: every ingredient of Weil’s scaffold is built (pairing, trace identity, effectivity, ample cone, the zero-carrying cohomology, the Frobenius flow, and an explicit $\mathcal{T}$ -independent $\mathfrak{sl}_2$), and the obstruction is isolated as the incompatibility of two specific structures. It does not assert that RH is false, nor that no proof exists — only that the hard-Lefschetz mechanism, as it operates in the function-field case, cannot be transported to ζ along this route.

8. CONCLUSION

The archimedean–modular construction produces three genuine gains: a positive metric $g_\infty = |\Psi|$ on the spectral bulk (Proposition 3.1); the first explicit hard-Lefschetz raising operator independent of the Frobenius (Proposition 5.2); and the exact transport identity (Proposition 6.1). It does not cross the wall: the Weil positivity is the near-cancellation residual $A_\infty - P$, and the Lefschetz dichotomy (Theorem 7.2) names precisely why the positivity cannot be supplied by a hard-Lefschetz class — linear independence of the zeros forbids the lowest-weight grading that Hodge–Riemann requires. The result is a structural theorem about the obstruction, consistent with the program’s wrong-sign map [2, 3], reached this time from the Hodge-theoretic side.

APPENDIX A. REPRODUCIBLE COMPUTATIONS

All values are produced by the following self-contained scripts (Python with mpmath/numpy); outputs are quoted in the text.

A.1 The archimedean density (Propositions 3.1).

```
import mpmath as mp
mp.mp.dps = 40
logpi = mp.log(mp.pi)
Psi = lambda r: mp.re(mp.digamma(mp.mpf('0.25')+1j*mp.mpf(r)/2)) - logpi
dPsi = lambda r: mp.re(0.5*1j*mp.polygamma(1, mp.mpf('0.25')+1j*mp.mpf(r)/2))
# (i) strict monotonicity, (ii) unique zero r0, (iii) asymptotics
print("Psi(0) =", mp.nstr(Psi(0),12)) # -5.37218341923 (global min)
print("r0 =", mp.nstr(mp.findroot(Psi,6.3),16)) # 6.289835988836903
print("gamma_1 =", mp.nstr(mp.im(mp.zetazero(1)),12)) # 14.1347251417 > r0
print("Psi(g1) =", mp.nstr(Psi(mp.im(mp.zetazero(1))),8)) # 0.81054874 > 0
for r in [20,100,1000]:
    print(r, mp.nstr(Psi(r)/(2*mp.pi),10), mp.nstr(mp.log(mp.mpf(r)/(2*mp.pi))/(2*mp.pi),10))
```

A.2 Indefiniteness on primitives (Proposition 4.1).

```
import mpmath as mp
mp.mp.dps = 30
logpi = mp.log(mp.pi)
Psi = lambda r: mp.re(mp.digamma(mp.mpf('0.25')+1j*mp.mpf(r)/2)) - logpi
def Ainf_over_norm(sig):
    sig=mp.mpf(sig)
    fh=lambda r:(r**2+mp.mpf('0.25'))*mp.e**(-r**2/(2*sig**2)) # primitive: fh(+i/2)=0
    A=(1/(2*mp.pi))*mp.quad(lambda r: fh(r)**2*Psi(r), [-mp.inf,-6.29,0,6.29,mp.inf])
    N=(1/(2*mp.pi))*mp.quad(lambda r: fh(r)**2, [-mp.inf,0,mp.inf])
    return A/N
for s in ['1','3','6','12']:
    print('sigma',s, mp.nstr(Ainf_over_norm(s),8)) # -1.82, -0.41, +0.30, +0.998
```

A.3 The continuum $s\ell_2$ (Proposition 5.2).

```
import numpy as np
N=6000; h=0.01; c=1.0; shift=int(round(c/h))
r=(np.arange(N)-N//2)*h
Tcont=np.diag(r)
Tc=np.zeros((N,N)); Tc[shift:,:-shift]=np.eye(N-shift) # (T_c f)_j=f_{j-shift}
Tmc=np.zeros((N,N)); Tmc[:,-shift,shift:]=np.eye(N-shift) # (T_c f)_j=f_{j+shift}
m=-(r**2)/c**2+r/c
H=(2.0/c)*Tcont; L=Tc; Lam=Tmc@np.diag(m)
comm=lambda A,B:A@B-B@A; s=slice(shift+20,N-shift-20); e=lambda M:np.max(np.abs(M[s,s]))
print(' [Tcont,L]-cL', e(comm(Tcont,L)-c*L)) # ~3e-15
print(' [H,L]-2L', e(comm(H,L)-2*L)) # ~7e-15
print(' [H,Lam]+2Lam', e(comm(H,Lam)+2*Lam)) # ~7e-12
print(' [L,Lam]-H', e(comm(L,Lam)-H)) # ~3e-13
```

A.4 Transport identity and the dichotomy comparison (Props. 6.1, 7.1).

```
import mpmath as mp
mp.mp.dps=25
logpi=mp.log(mp.pi)
Psi=lambda r: mp.re(mp.digamma(mp.mpf('0.25')+1j*mp.mpf(r)/2))-logpi
def vonmangoldt(nmax):
    spf=list(range(nmax+1)); i=2
    while i*i<=nmax:
        if spf[i]==i:
            for j in range(i*i,nmax+1,i):
                if spf[j]==j: spf[j]=i
        i+=1
```

```

L=[mp.mpf(0)]*(nmax+1)
for n in range(2,nmax+1):
    p=spf[n]; m=n
    while m%p==0: m//=p
    if m==1: L[n]=mp.log(p)
return L
NMAX=20000; Lam=vonmangoldt(NMAX); Z=[mp.im(mp.zetazero(k)) for k in range(1,301)]
for s in ['2','3','4']:
    sig=mp.mpf(s); h=lambda r:(r**2+mp.mpf('0.25'))**2*mp.e**(-r**2/sig**2)
    g=lambda u:(1/(2*mp.pi))*mp.quad(lambda r:h(r)*mp.cos(u*r),[-mp.inf,0,mp.inf])
    Q=mp.fsum(2*h(z) for z in Z)
    A=(1/(2*mp.pi))*mp.quad(lambda r:h(r)*Psi(r),[-mp.inf,-6.29,0,6.29,mp.inf])
    gI=(1/(2*mp.pi))*mp.quad(lambda r:h(r)*abs(Psi(r)),[-mp.inf,-6.29,0,6.29,mp.inf])
    P=mp.fsum(2*(Lam[n]/mp.sqrt(n))*g(mp.log(n)) for n in range(2,NMAX+1) if Lam[n]!=0)
    print(s,'Q',mp.nstr(Q,6),'A',mp.nstr(A,6),'P',mp.nstr(P,6),
          'Q-(A-P)',mp.nstr(Q-(A-P),4),' g_inf',mp.nstr(gI,6))
# g_inf >= 0 but differs from Q by 10^-2..10^-18: the grading yields g_inf, not Q.

```

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